



29/8/26

SIMULATION OF AUTOCORRELATION AND PSD:-

AIM:- To generate and analyse a Auto correlation and the PSD by using the scilab.

SOFTWARE:-
scilab software

CODE:-

```
Am = 2;  
Ac = 4;  
fm = 20;  
fc = 200;  
fs = 2000;  
t = 0:1/fs : 0.05;  
x = Am * cos(2*pi*fm*t);  
subplot(3,2,1);  
plot(t,x);  
title("Input cosine signal");  
xlabel("Time");  
ylabel("Amplitude");  
au = x(orr(x,x));  
subplot(3,2,2);  
plot(au);  
title("Auto-correlation");  
xlabel("Lag");  
ylabel("Correlation");
```

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1891-1892

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Scilab Program Implementation (20)	Waveform and tabulation (20)	calculation and result (20)	viva- voce (20)	Result & observation (20)
20	20	20	15	20 95

RESULT:-

The Auto correlation and Power spectral Density (PSD) of the given cosine wave were successfully simulated and plotted using Scilab.